

KONGU ENGINEERING COLLEGE

SMART IRRIGATION AND WATER MANAGEMENT SYSTEM MECHATRONICS AND IOT

EXCERSICE - 01

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SMART IRRIGATION AND WATER MANAGEMENT SYSTEM

1. Project Overview

Introduction

Agriculture requires a significant amount of water, and conventional irrigation methods often result in unnecessary water consumption due to over-irrigation or improper scheduling. The Smart Irrigation and Water Management System is an IoT-based automated irrigation system designed to provide water to plants according to the moisture condition of the soil.

The system uses a soil moisture sensor to continuously monitor the moisture level of the soil. An ESP8266 NodeMCU acts as the main controller and processes the sensor information. A DHT11 sensor is also used to measure environmental temperature and humidity. Based on the soil moisture level, the controller automatically operates a relay, which switches the water pump ON or OFF.

The main objective is to reduce water wastage, minimize manual intervention, and provide efficient irrigation to plants.

Main Features

- Automatic irrigation based on soil moisture.
- Continuous soil moisture monitoring.
- Temperature and humidity monitoring using DHT11.
- Automatic water pump control.
- Relay-based pump switching.
- Reduced water wastage.
- Reduced manual intervention.
- IoT-ready system using ESP8266.

2. Problem Statement

Traditional irrigation systems commonly depend on manual operation or fixed irrigation schedules. These methods do not consider the actual moisture condition of the soil.

This can result in:

- Excessive water usage.
- Under-irrigation of plants.
- Over-irrigation and waterlogging.

- Unnecessary operation of water pumps.
- Increased electricity consumption.
- Requirement for continuous human monitoring.

Therefore, an automated system is required to monitor soil conditions and provide water only when it is necessary.

3. Proposed Solution

The proposed system uses sensors and a microcontroller to automate irrigation.

The soil moisture sensor measures the moisture content of the soil and sends the sensor value to the ESP8266 NodeMCU. The DHT11 sensor measures temperature and humidity.

The ESP8266 processes these values and determines whether irrigation is required.

When the soil becomes dry:

Soil Moisture Sensor → ESP8266 → Relay ON → Water Pump ON

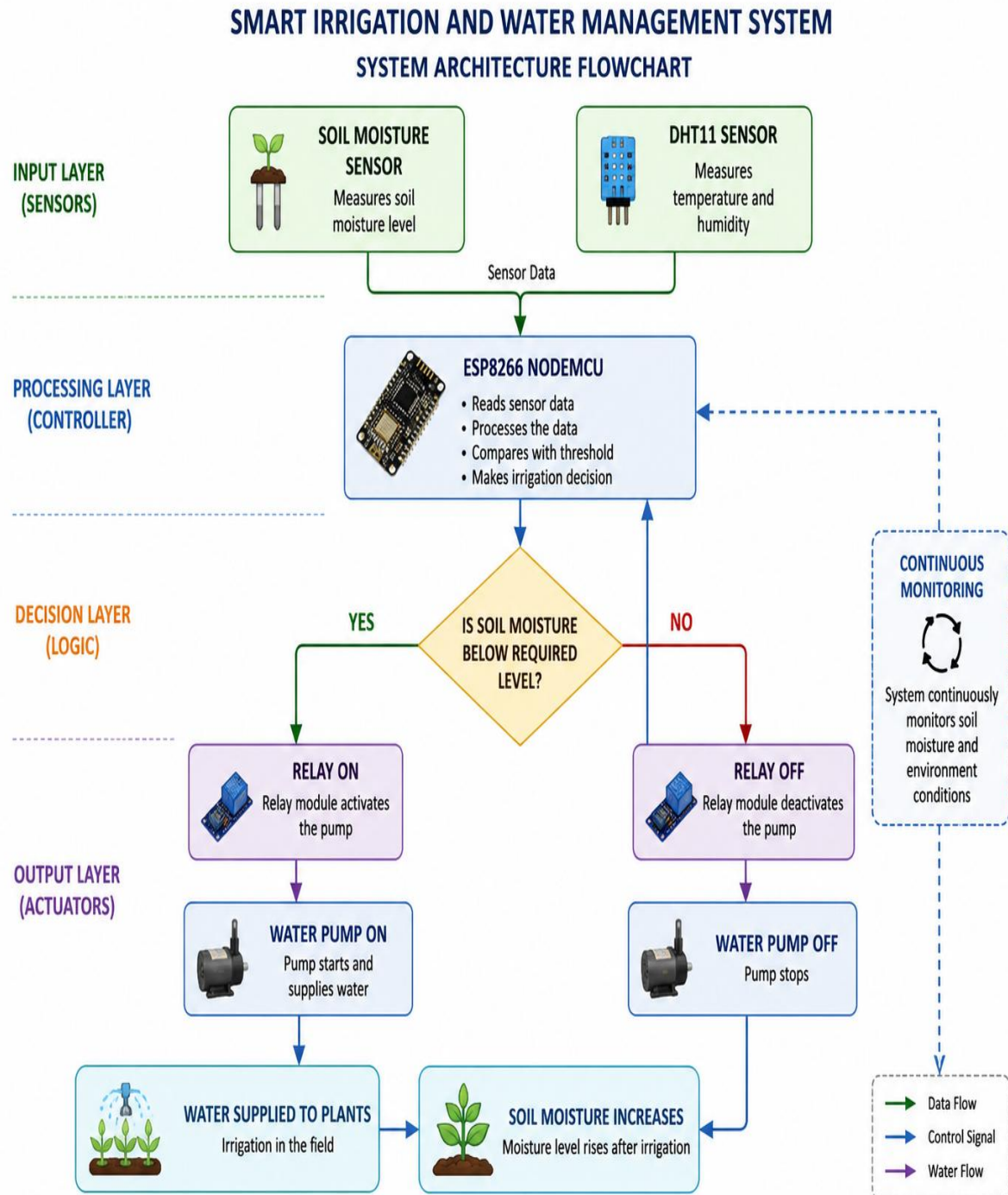
When sufficient moisture is reached:

Soil Moisture Sensor → ESP8266 → Relay OFF → Water Pump OFF

This automatic control helps provide water according to the requirement of the plants.

4. System Architecture

The overall architecture of the system is:



System Flow

Sensors → Microcontroller → Decision Making → Relay → Pump → Irrigation

5. Circuit Table

Component	Pin	ESP8266 NodeMCU
Soil Moisture Sensor	VCC	3V3
Soil Moisture Sensor	GND	GND
Soil Moisture Sensor	AO	A0
DHT11	VCC	3V3
DHT11	GND	GND
DHT11	DATA	D2
Relay Module	VCC	VIN
Relay Module	GND	GND
Relay Module	IN	D3
Relay Module	COM	Pump power supply +
Relay Module	NO	Pump +
Water Pump	–	Pump power supply –

Power Supply

The ESP8266 can be powered through USB. The relay is powered from the appropriate 5V supply through VIN when the board is USB powered.

The water pump should use a suitable power supply according to its rated voltage and current. For a 5V pump, a regulated 5V supply or suitable boost converter should be used.

6. Circuit Design

The circuit consists of five major sections:

6.1 ESP8266 NodeMCU

The ESP8266 is the main controller. It receives sensor data and controls the relay according to the soil moisture condition.

6.2 Soil Moisture Sensor

The soil moisture sensor is connected to the analog input A0.

It provides an analog value representing the moisture condition of the soil.

Soil Sensor AO → ESP8266 A0

6.3 DHT11 Sensor

The DHT11 measures:

- Temperature.
- Relative humidity.

DHT11 DATA → ESP8266 D2

6.4 Relay Module

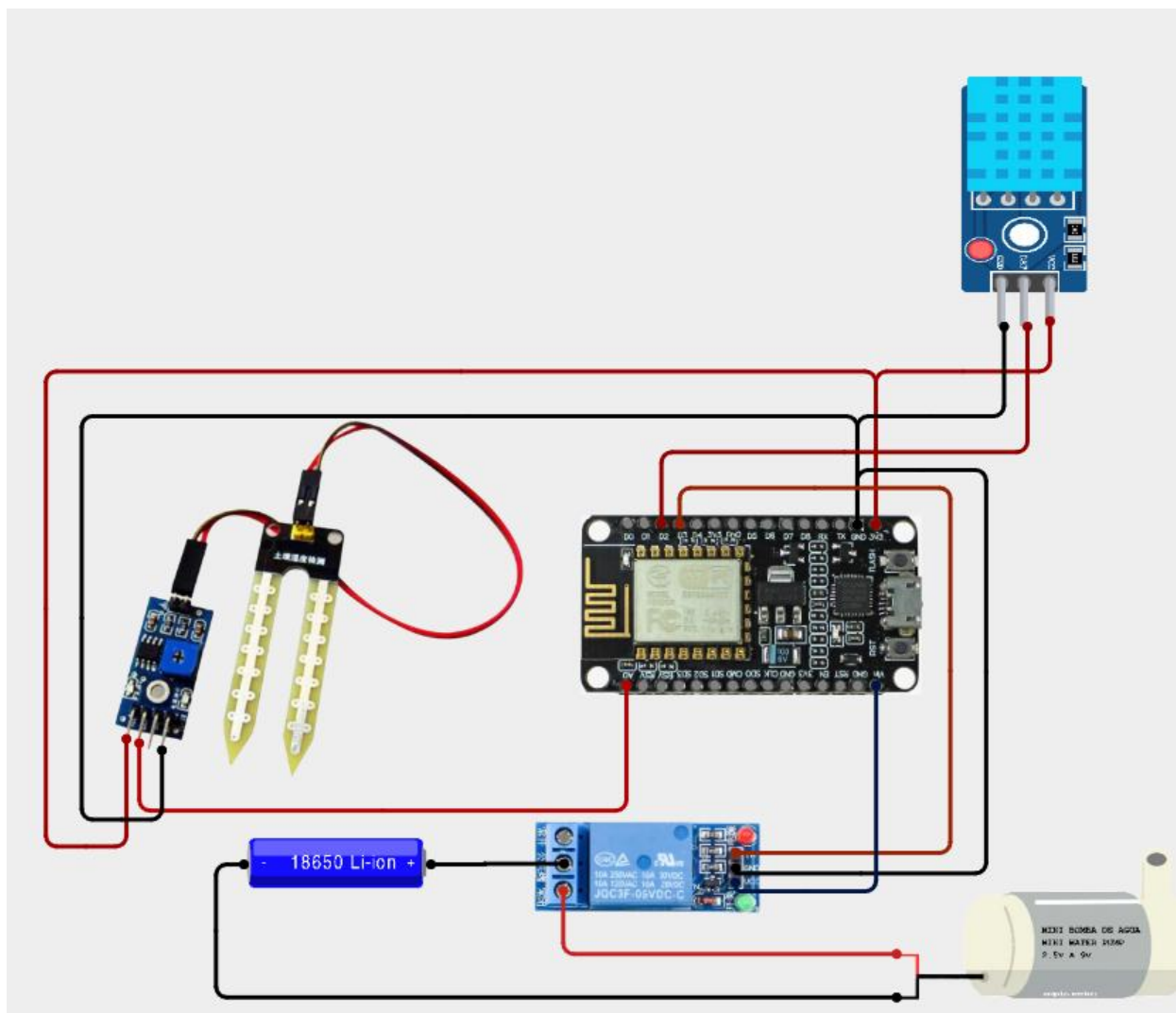
The relay provides electrical switching between the low-power controller and the water pump.

Relay IN → ESP8266 D3

6.5 Water Pump

The water pump supplies water to the plants. It is controlled through the relay and should not be connected directly to an ESP8266 GPIO pin.

CIRCUIT DESIGN:



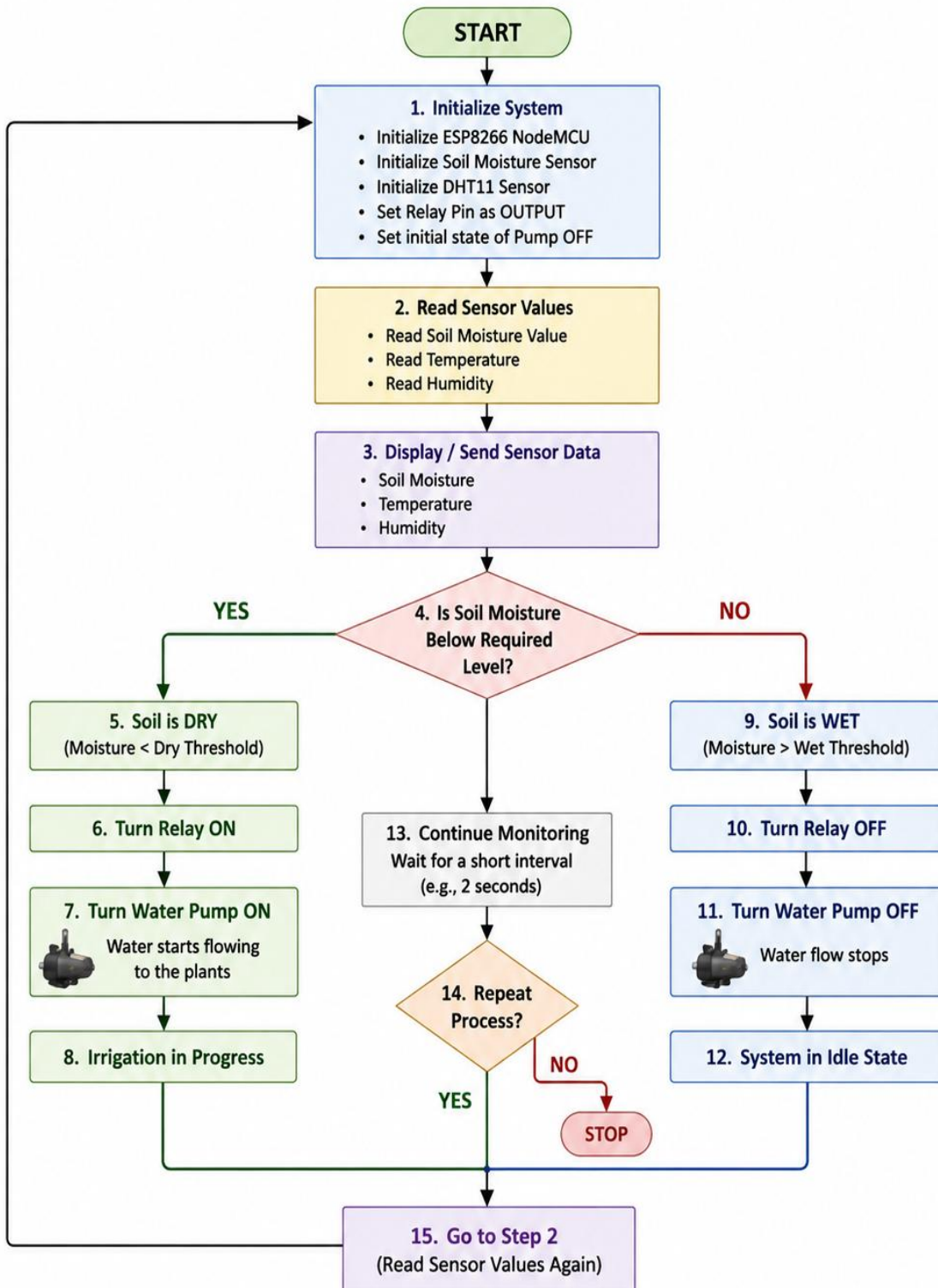
7. Algorithm

Algorithm

1. Start the system.
2. Initialize the ESP8266.
3. Initialize the soil moisture sensor.
4. Initialize the DHT11 sensor.
5. Configure the relay as an output.
6. Read the soil moisture value.
7. Read temperature and humidity.
8. Compare the soil moisture value with the predefined threshold.
9. If the soil is dry, turn ON the relay.
10. The relay turns ON the water pump.
11. Continue monitoring the soil.
12. When sufficient moisture is reached, turn OFF the relay.
13. The water pump stops.
14. Display or transmit the sensor readings.
15. Repeat the process continuously.

SMART IRRIGATION AND WATER MANAGEMENT SYSTEM

ALGORITHM FLOWCHART



LEGEND



8. Coding

The following Arduino program is designed for the ESP8266 NodeMCU used in the project.

```
#include <DHT.h>

#define SOIL_PIN A0
#define DHT_PIN D2
#define RELAY_PIN D3

#define DHT_TYPE DHT11

DHT dht(DHT_PIN, DHT_TYPE);

int dryThreshold = 700;
int wetThreshold = 500;

void setup() {

    Serial.begin(115200);

    pinMode(SOIL_PIN, INPUT);
    pinMode(RELAY_PIN, OUTPUT);

    // Pump OFF initially
    digitalWrite(RELAY_PIN, HIGH);

    dht.begin();

    Serial.println("Smart Irrigation System Started");
}
```

```
void loop() {

    int soilValue = analogRead(SOIL_PIN);

    float temperature = dht.readTemperature();
    float humidity = dht.readHumidity();

    Serial.println("-----");

    Serial.print("Soil Moisture: ");
    Serial.println(soilValue);

    if (isnan(temperature) || isnan(humidity)) {

        Serial.println("DHT11 reading failed");

    } else {

        Serial.print("Temperature: ");
        Serial.print(temperature);
        Serial.println(" C");

        Serial.print("Humidity: ");
        Serial.print(humidity);
        Serial.println(" %");
    }

    if (soilValue >= dryThreshold) {
```

```
digitalWrite(RELAY_PIN, LOW);

Serial.println("Soil Status: DRY");
Serial.println("Pump Status: ON");

}

else if (soilValue <= wetThreshold) {

digitalWrite(RELAY_PIN, HIGH);

Serial.println("Soil Status: WET");
Serial.println("Pump Status: OFF");

}

else {

Serial.println("Soil Status: MODERATE");
Serial.println("Pump Status: Previous State");

}

delay(2000);
}
```

Code Explanation

- A0 reads the soil moisture sensor.
- D2 receives data from the DHT11.
- D3 controls the relay.
- dryThreshold determines when irrigation should start.
- wetThreshold determines when irrigation should stop.
- The DHT11 provides temperature and humidity information.
- The relay controls the water pump.

The threshold values should be calibrated according to the actual soil and sensor used.

9. System Working

When the system is powered ON, the ESP8266 initializes all connected sensors and the relay.

The soil moisture sensor continuously checks the moisture condition of the soil.

Dry Soil Condition

When the soil moisture value indicates that the soil is dry:

Soil Dry

↓

ESP8266 detects dry condition

↓

Relay ON

↓

Water Pump ON

↓

Water supplied to plants

Wet Soil Condition

When the soil reaches the required moisture level:

Soil Wet

↓

ESP8266 detects sufficient moisture

↓

Relay OFF

↓

Water Pump OFF

At the same time, the DHT11 continuously measures temperature and humidity.

This process repeats automatically, allowing the system to maintain suitable soil moisture.

10. Bill of Materials

S.No.	Component	Quantity
1	ESP8266 NodeMCU	1
2	Soil Moisture Sensor	1
3	1-Channel Relay Module	1
4	Mini Water Pump	1
5	DHT11 Sensor	1
6	Breadboard	1
7	Jumper Wires	1 set
8	Suitable Pump Power Supply	1

Software Requirements

- Arduino IDE
- ESP8266 Board Package
- DHT Sensor Library
- Cirkuit Designer for circuit design/simulation

11. Prototype Implementation

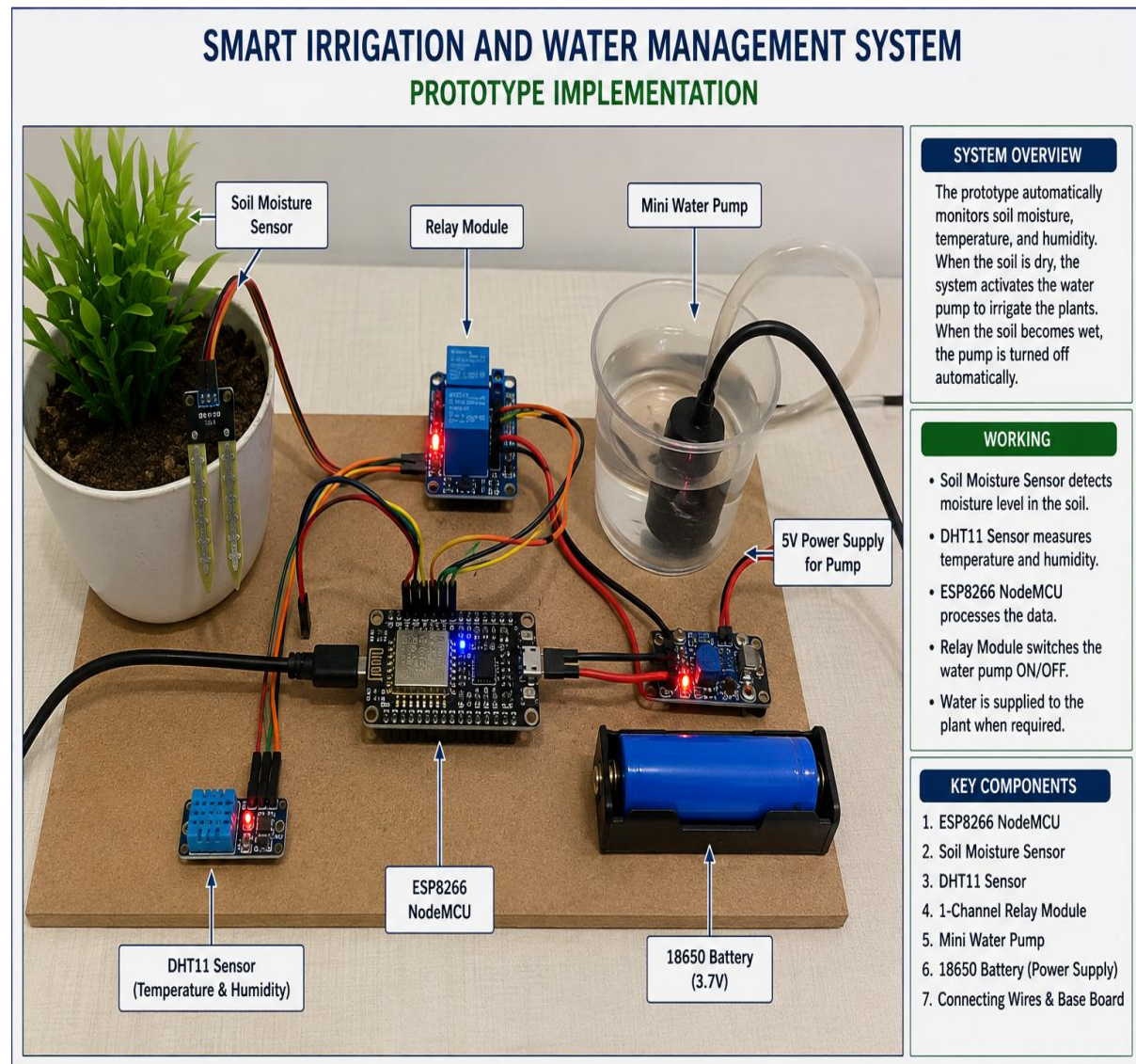
The prototype was designed using an ESP8266 NodeMCU, soil moisture sensor, DHT11 sensor, relay module and mini water pump.

The components were first arranged and connected in Cirkuit Designer to develop the circuit design.

The ESP8266 was selected as the main controller because it provides sufficient GPIO and analog input capability and can also support Wi-Fi-based IoT extensions.

The prototype operates automatically according to the soil moisture condition.

Prototype Process



Simulation

The circuit can be represented in Circuit Designer. For simulation, a supported simulation microcontroller such as Arduino Uno can be used if the ESP8266 NodeMCU configuration does not provide an active simulation/verification environment.

12. Results & Performance Analysis

Expected Results

The system successfully demonstrates automatic irrigation based on soil moisture.

Condition	Sensor Condition	Relay	Pump
Dry soil	Moisture below required level	ON	ON
Moderate soil	Intermediate value	Maintains previous state	Maintains previous state
Wet soil	Moisture sufficient	OFF	OFF

Performance

The system provides the following benefits:

- Automatic irrigation.
- Reduced unnecessary water usage.
- Reduced manual intervention.
- Prevention of over-irrigation.
- Continuous environmental monitoring.
- Automatic pump control.
- Potential for remote IoT monitoring

Conclusion

The Smart Irrigation and Water Management System provides an efficient approach to automated agricultural irrigation. By continuously monitoring soil moisture and environmental conditions, the system can automatically operate the water pump only when irrigation is required.

The ESP8266 acts as the central controller, while the soil moisture sensor provides information about the condition of the soil. The DHT11 provides temperature and humidity measurements, and the relay controls the water pump.

The proposed system can help reduce water wastage and improve irrigation efficiency. In future development, additional features such as water-level monitoring, flow measurement, mobile applications, cloud data logging, weather-based irrigation and solar-powered operation can be incorporated.